

Lump Sum vs DCA Calculator

Personal Financial Scenario Report

United States · USD

fincalcsmart.com/lump-sum-vs-dca-calculator

EXECUTIVE SUMMARY

Similar Outcomes

\$59,914	\$58,906	\$1,008	28/100	Similar
Lump Sum FV	Monthly DCA FV	Lump Sum Advantage	Advantage Score Minimal	Comparison

FINAL VALUE COMPARISON (LUMP SUM VS DCA)



Total projected: \$118,820

Lump Sum FV: \$59,914 (50.4%)

Monthly DCA FV: \$58,906 (49.6%)

AI-ASSISTED LUMP SUM VS DCA SUMMARY

Investing \$25,000 at 6% (Monthly compounding) over 15 years: lump sum grows to \$59,914 (gain: \$34,914, +140%). Monthly strategy (\$4,167/mo over 6 months) reaches \$58,906 (gain: \$33,906, +136%). Lump Sum Advantage Score: 28/100 (Minimal).

Lump sum advantage: \$1,008 more at year 15 (minimal difference). Milestone comparison -- at year 10: lump \$44,771 vs DCA \$44,018; at year 20: \$80,178 vs \$78,830; at year 30: \$143,587 vs \$141,172.

Lump sum investing tends to outperform gradual deployment in rising markets because the full principal compounds from day one. The monthly strategy (DCA) spreads market entry over 6 months, reducing timing risk but also reducing time-in-market for early tranches. Research consistently shows lump sum outperforms DCA approximately two-thirds of the time in equity markets. However, DCA is often the psychologically easier approach for risk-averse investors or when capital is received gradually (e.g., payroll). This model does not account for market volatility -- the advantage of DCA in volatile or declining markets can be larger than this comparison suggests.

INPUTS & ASSUMPTIONS

Total Investment	\$25,000
Annual Return Rate	6%
Compounding Frequency	Monthly
Investment Horizon	15 years
DCA Spread Period	6 months
Monthly DCA Amount	\$4,166.67
Region	United States

DETAILED RESULTS

Lump Sum: Final Value	\$59,914
Lump Sum: Gain	\$34,914
DCA: Final Value	\$58,906
DCA: Gain	\$33,906
Lump Sum Advantage	\$1,008
Difference Significant	No
Advantage Score	28/100 (Minimal)
At Year 10: Lump Sum	\$44,771
At Year 10: DCA	\$44,018
At Year 20: Lump Sum	\$80,178
At Year 20: DCA	\$78,830
At Year 30: Lump Sum	\$143,587
At Year 30: DCA	\$141,172

KEY DRIVERS / POSSIBLE ADJUSTMENTS

- The 6% return rate is the primary driver of the advantage gap. At higher return rates, lump sum benefits more from full compounding from day one -- the advantage widens with higher assumed returns.
- Spread period of 6 months directly determines the DCA disadvantage. A 3-month spread leaves most capital at risk for almost the full horizon; a 36-month spread more meaningfully reduces market-timing risk.
- Total investment horizon of 15 years amplifies any early-compounding advantage. At longer horizons, even a small initial advantage compounds into a larger absolute gap.

METHODOLOGY

What this calculator models:

- Lump sum: full principal P compounded for T months using the effective monthly rate derived from the selected compounding frequency: $FV = P \times (1 + r)^T$.
- Monthly strategy (DCA): equal monthly instalments of $\text{totalAmount} / \text{spreadMonths}$, each compounded from its contribution month to end of horizon. $FV = \text{sum of monthly } x \times (1+r)^{(T-k)} \text{ for } k = 1 \text{ to } \text{spreadMonths}$.
- Advantage Score (0-100): derived from the percentage gain difference between strategies. Significant threshold: >5% absolute gain difference. ComparisonState "similar" when gainDelta is below the significance threshold.

What this calculator does not model:

- Market volatility -- lump sum outperforms in rising markets; DCA reduces loss in declining markets.
- Transaction costs, fund management fees (MER), or tax on capital gains.
- Dividend reinvestment or yield on fixed-income components.
- Tax-sheltered account rules (TFSA/RRSP in CA, 401k/IRA in US) that affect net return.
- Behavioural factors -- DCA often leads to better investor behaviour during market downturns.
- Sequencing: this model assumes the full lump sum is available immediately.

EDUCATIONAL DISCLAIMER

This report is for educational and illustrative purposes only. Investment return rates and compounding frequencies are assumptions entered by the user and do not represent guaranteed returns. Past market performance does not predict future results. This does not constitute investment advice. Consult a registered investment advisor before deploying capital.